Cumulative probability at Jkn: pk = 11/12
Maximum absolute damage length: lmax = 60 m
Length where normalized distribution ends: L* = 260 m

Probability density at J = 0:

  = knkknk J JpJpb max0 12

When :

=
s
m
L
lJ J maxmax , min

()
0
2 2
00
4
12 1 1 1
2 b

---CHUNK---

SOURCE: MSC.194(80).pdf
Length where normalized distribution ends: L* = 260 m

Probability density at J = 0:

  = knkknk J JpJpb max0 12

When :

=
s
m
L
lJ J maxmax , min

()

0
2 2
00
4
12 1 1 1
2 b
J b J b pJJ mm kmk + + + =

0 12b b=

When :
= *maxmax* , minLlJJm

()
0
2* 2
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12 1 1 1
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J b J b pJJ mm kmk + + + =

s
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L JJ ** =

s
k
k
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* *
=

= k mkkk J JpJpb 1212

*L Ls
*L Ls >
RESOLUTION MSC.194(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-
20 -

() 211 214 kkk k mk JpJ J Jpb =

()
221
12
k m
k
J J
pb

=

mJ b b2122 =

The non-dimensional damage length:

()
sL
1 x 2 xJ =
The normalized length of a compartment or group of compartments:

Jn is to be taken as the lesser of J and Jm

1.1.1 Where neither limits of the comp artment or group of compartments under consideration coincides with the aft or forward terminals:

J Jk :
() 121121 3b J b J61) , (+ = =p 2 x 1 x p

J>Jk :
() ()
3 3
2112
2
1211
3
112
3
1
2
1
3

---CHUNK---

SOURCE: MSC.194(80).pdf
1.1.1 Where neither limits of the comp artment or group of compartments under consideration coincides with the aft or forward terminals:

J Jk :
() 121121 3b J b J61) , (+ = =p 2 x 1 x p

J>Jk :
() ()
3 3
2112

2
1211
3
112
3
1
2
1
3
1) , (k nkkk J J b JJ b J b J b J b p 2 x 1 x p + + = =
() ( ) () k nk n J J J b J J b J b + +22
2 2
2221
2
1

1.1.2 Where the aft limit of the compar tment or group of compartments under consideration coincides with the aft terminal or the forward limit of the compartment or group of compartments under consideration coincides with the forward terminal:

J Jk:
() J p 2 x 1 x p+ =1
2